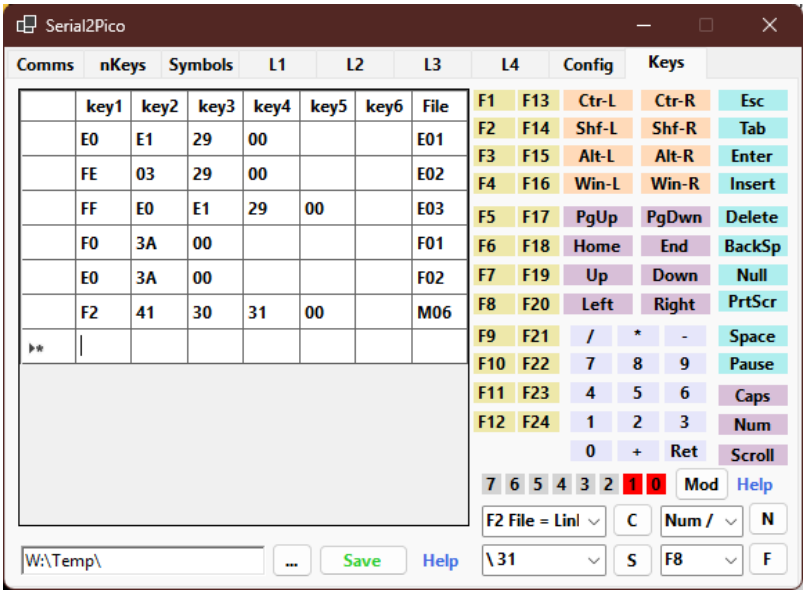
Exercise in using the Keys and Link tabs in the PC App

Construct the following 6 files in the Keys tab of the PC App – press Show Macro Keys in the Config Tab



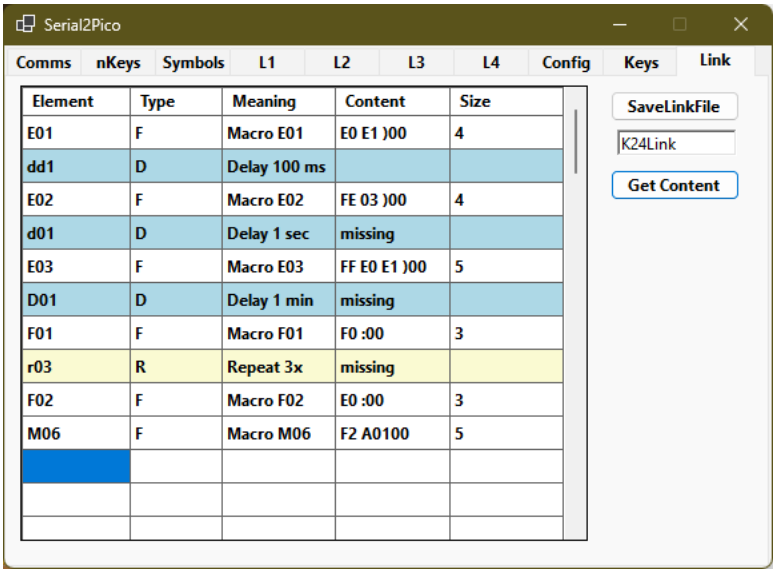
For file E01: Click in first cell of first row (below key1), then click Ctr-L, then click Shf-L, then Esc, then Null. Type E01 into the cell below File.
For file E02: Click on the 0 and 1 blocks of the Mo(d) list – they will turn red – that means Ctrl-L and Shf-L are on. Then click on the first cell of the second row and open the list next to the [C] button and select FE 1x Mod byte, then press the button [C] and check if FE shows in the first cell. Then click on the [Mod] button with the two red [1][0] labels next to it and check if 03 is entered in the cell next to FE. Then click on Esc, then Null and name the file E02
For File E03: Click on the first cell of the third row and open the list next to the [C] button and select FF Mod byte, then press the button [C] and check if FF shows in the first cell. Then click Ctr-L, then click Shf-L, then Esc, then Null. Type E03 into the cell below File.
Explanation: File E01 presses the keys Control + Shift + Esc together which opens the Task Manager in Windows. File E01 does the same action but uses the modifier bit to indicate that Control + Shift are pressed, namely the value 03 which is 0000 0011, then it adds the Esc key value 29 – it uses the flag FE to indicate that the next value 03 is the modifier byte. File E03 also the same action but uses the flag FF to indicate the modifier bit must be constructed from the next two values for the Control and Shift keys, then it also adds the Esc key value 29.

For File F01: Click on the first cell of the fourth row and open the list next to the [C] button and select F0 F1-F24, then press the button [C] and check if F0 shows in the first cell. Open the list next to the [F] button and select F1, then press the button [F] and check if 3A shows in the second cell. Then click on the Null label. Type F01 into the cell below File.
For File F02: Click on the first cell of the fifth row then click on the label Shf-L. Open the list next to the [F] button and select F1, then press the button [F] and check if 3A shows in the second cell. Then click on the Null label. Type F02 into the cell below File.
Explanation: You only need the flag F0 if the first command is F1-F24. If it is a 2nd or 3rd command you only need the F1-F24 value without the F0.

For File M06: Click on the first cell of the sixth row and open the list next to the [C] button and select F2 File=Link, then press the button [C] and check if F2 shows in the first cell. Open the list next to the [F] button and select F8 and press [F], then open the list next to the [S] button and select] 30 and press [S], then and select \ 31 and press [S]. press the label Null. Check that the row has the values F2 41 30 31 00.
Explanation: The flag F2 indicates that the three character filename after the F2 should not be executed but that is contains another filename that must be executed – i.e. it is a symbolic link file. Because that filename must be in ASCII letters – it is A01 – and the keyboard is operating in HID mode in the Keys tab of the PS App, you need to enter the ASCII letter A01 by using the F8=41=A,]=30=0, and /=31=1 to form the filename A01.

You can now save the six files by pressing the button [Save] – if the path next to the button is empty first select a folder where the files must be saved using the [...] button. Check in the folder if the six files E01, E02, E03, F01, F02, M06 are there – if you have a hex editor, you can check their content as well.

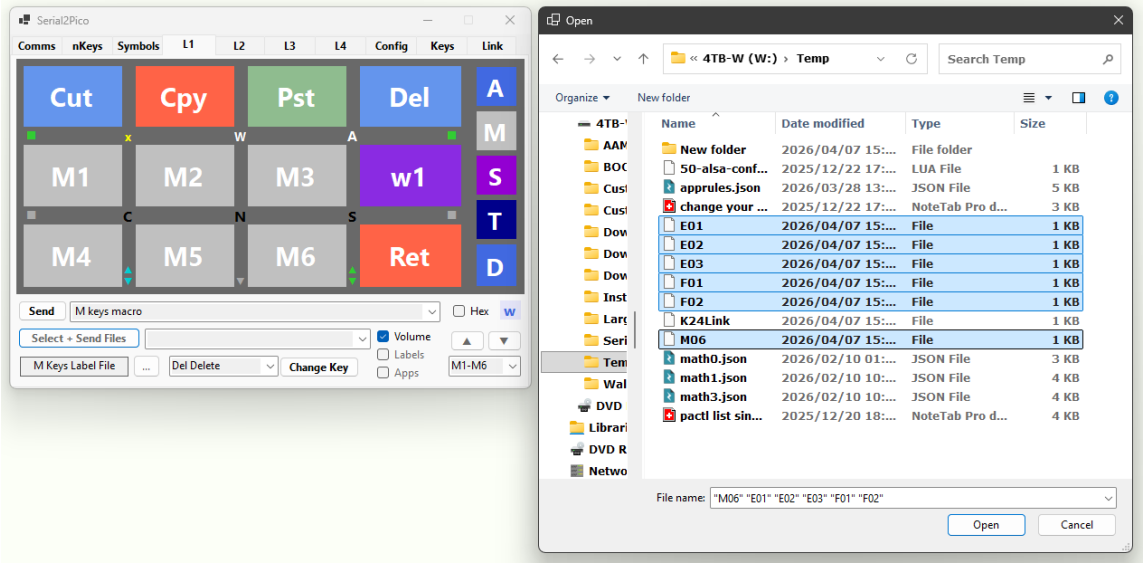
Now open the tab Link by clicking the Show Link List in the Config tab:



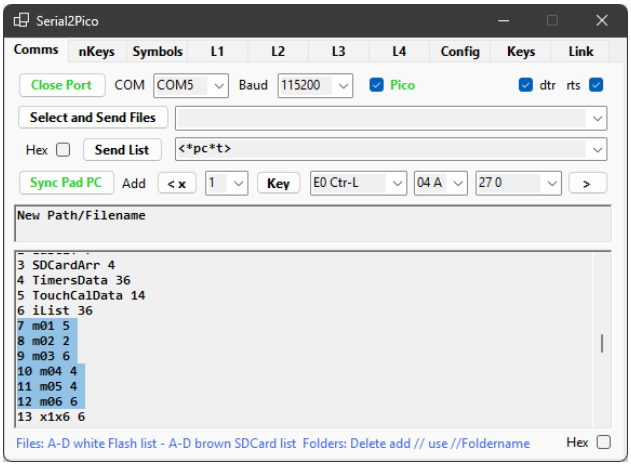
Element	Type	Meaning	Content	Size
E01	F	Macro E01	E0 E1)00	4
dd1	D	Delay 100 ms		
E02	F	Macro E02	FE 03)00	4
d01	D	Delay 1 sec	missing	
E03	F	Macro E03	FF E0 E1)00	5
D01	D	Delay 1 min	missing	
F01	F	Macro F01	F0 :00	3
r03	R	Repeat 3x	missing	
F02	F	Macro F02	E0 :00	3
M06	F	Macro M06	F2 A0100	5

When first opened it will not contain anything – start by typing in the first filename E01 in the first row below Element, then type in D01, the third row has the file E02, then type d99, then type E03, DD1, F01, D01, r05, F02, M06. Then press the [Get Content] button and check if the values are the same as what is shown above. Note that file M06 has the filename A01 in its content. Type in a Link filename such as K24Link and press [SaveLinkFile] which will save the file as K24Link which is a plain text file with the content E01dd1E02d01E03D01F01r03F02M06.
Explanation: The file K24Link will execute the 3-character commands one after the other. First file E01, then it will wait for I00 milliseconds, then execute file E02 and wait for 1 second, and the type of actions further on in the list. The repeat r03 means it will do the action in file M02 three times (it opens and closed the Word ribbon).

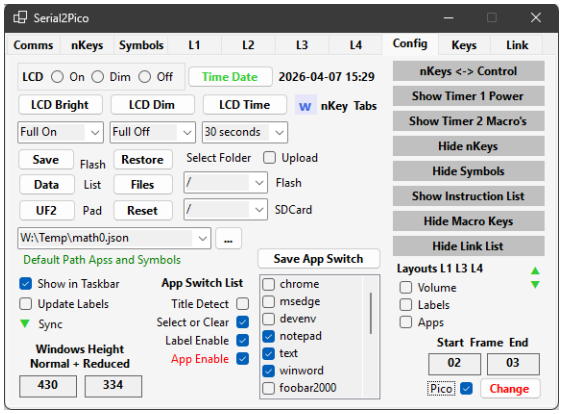
You can now upload the files to the Pico macro-pad:

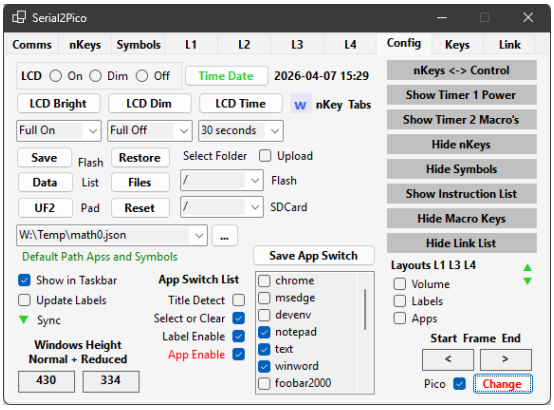


Open L1 – the M keys and press [Select + Send Files], open the folder where your six files are located and select them then press [Open] and the files will be copied as m01 to m06 to either the SDCard if you have a brown A-D or Flash if it is a white A-D.

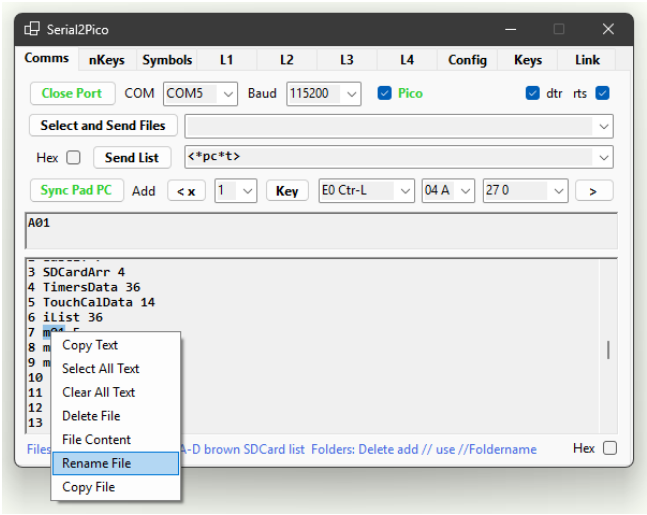


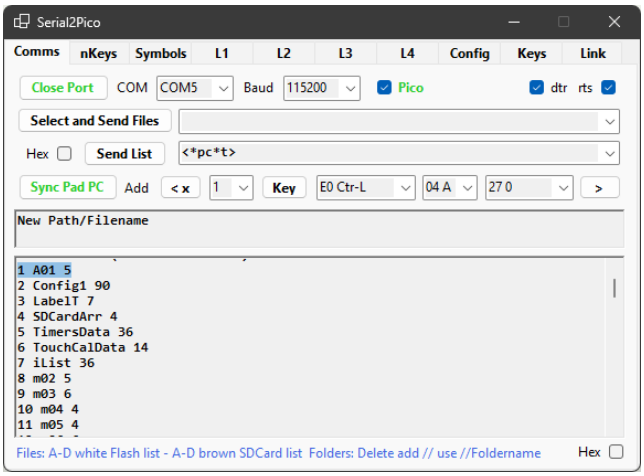
You will notice that saving the files have added one null byte at the end but importantly, file m02 which is file E02 on the PC has been truncated to 2 bytes. The reason is that the macropad and PC App, are using framing stop and start bytes of 02 and 03, therefore the Pico saw the 03 modifier byte and used it as a stop byte to truncate and save the file. To get around this for file E02 change the stop and start bytes to < > in the Config tab (check the Pico box for confirmation):

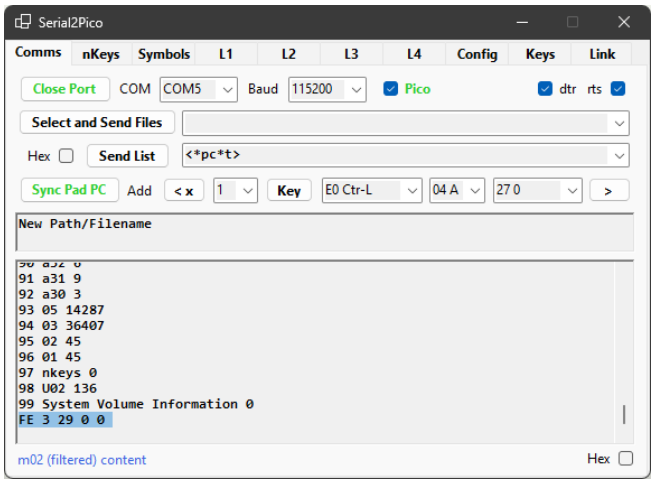


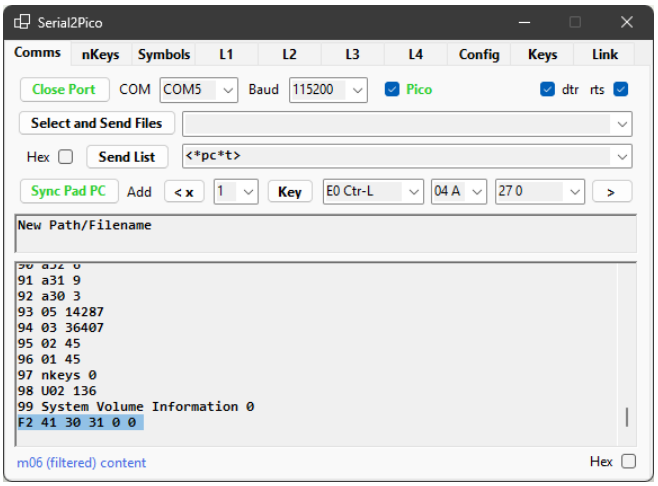


Then press change, and re-upload the file E02 which will get the name m02. Now make sure you are in the same storage as where your six files are saved – if Flash the white A-D, if SD Card then brown -A-D. You can now test the files by pressing keys m01, m02, m03, m04 and m05. For m04, and m05 have word open when testing the keys. For key m06 to work it needs a file A01 – I renamed file m01 to A01 by typing the name A01 in the textbox above the file list, then right-click and select rename. Then you can press m06 - it should execute the contents of file A01 which is to open the Task Manager. Remember that the Flash filesystem is case-sensitive so files a01 and A01 are two different files – this is not the case for the SD Card.






Serial2Pico
Comms nKeys Symbols L1 L2 L3 L4 Config Keys Link
Close Port COM COM5 Baud 115200 Pico dtr rts
Select and Send Files
Hex Send List <*pc*t>
Sync Pad PC Add < x 1 Key E0 Ctr-L 04 A 27 0 >
New Path/Filename
91 a31 9
92 a30 3
93 05 14287
94 03 36407
95 02 45
96 01 45
97 nkeys 0
98 U02 136
99 System Volume Information 0
FE 3 29 0 0
m02 (filtered) content
Hex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Sync Pad PC Add < x 1 Key E0 Ctr-L 04 A 27 0 >
New Path/Filename
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92 a30 3
93 05 14287
94 03 36407
95 02 45
96 01 45
97 nkeys 0
98 U02 136
99 System Volume Information 0
F2 41 30 31 0 0
m06 (filtered) content
Hex